

R1 — ML Research Track

Member: _____

Your role

You own the baseline arm (seed 0) of the reproduction. You write the final verdict on the perplexity claim.

Weekly tasks

Week 1 — Setup and data

- Task: Clone the repo. Set up the Python env with `uv sync`. Run the data prep on Colab: `uv run python -m research.tinystories.prepare --vocab 32768`.
- Deliverable: The folder `data/tinystories/vocab-32768/` exists with `tokenizer.json` and the token bins.
- Verify: The command finishes without error. The `tokenizer.json` shows 32768 tokens.
- Learn: Neural network basics. Watch the 3Blue1Brown series on neural networks. Write a small MLP in numpy that learns XOR.

Week 2 — Launch baseline seed 0

- Task: Launch the run: `uv run python -m research.tinystories.train --arm baseline --vocab 32768 --steps 3000 --target-core 559000 --seed 0` on your Colab account.
- Deliverable: The run reaches 3000 steps. The checkpoint and the run JSON record land in `runs/`.
- Verify: Checkpoints save to Google Drive every 1000 steps. The record shows the exact hyperparameters.
- Learn: Embeddings and next-token prediction. Explain what the tokenizer does and why the vocab is 32768.

Week 3 — Collect baseline results

- Task: Collect your two baseline checkpoints. Record the val ppl from the run JSON records for both seeds.
- Deliverable: The baseline ppl numbers for both seeds.
- Verify: Your numbers sit within ± 0.05 of 12.58.
- Learn: The transformer block. Understand LayerNorm, residuals, and the MLP inside one block.

Week 4 — Verdict on the perplexity claim

- Task: Run `uv run python -m research.tinystories.analyze --tag cleandeploy --expect-arms baseline,ple,fatembed --expect-seeds 2`. Write the verdict block for the verification report.
- Deliverable: The official ppl table and the verdict. Does the $+0.098$ nats edge reproduce?
- Verify: Your verdict matches the numbers in the tracking sheet.
- Learn: Loss, perplexity, and evaluation. Derive why $\text{ppl} = \exp(\text{loss})$.

Week 5 — PLE paper study

- Task: Lead the study of the Gemma 3n per-layer embeddings paper. Present it at the weekly meetup.
- Deliverable: A 1-page explainer of PLE. Why does the table live in flash? Why does vocab size matter?
- Verify: The team can answer: where does the 25M table live and why?
- Learn: Attention math. Write the QKV attention as numpy code.

Week 6 — Fine-tune the domain model

- Task: Lead the fine-tune run on the domain dataset from H3.
- Deliverable: The fine-tuned checkpoint and the ppl before/after table.
- Verify: The ppl on domain validation data improves after fine-tuning.

- Learn: Fine-tuning and transfer learning.

Week 7 — Teach attention

- Task: Present the attention concept to the team at the meetup.
- Deliverable: Your teach-back slides.
- Verify: H3 can explain attention back to you in one sentence.
- Learn: Prepare your presentation deeply. Teaching is learning.

Week 8 — Final report

- Task: Write the final report: verification results plus improvise results.
- Deliverable: FINAL_REPORT.md in the club repo.
- Verify: Every claim row in the tracking sheet has a verdict.
- Learn: Write the "what I learned" page. One page, honest.

Learning path

1. Neural network basics: perceptron, MLP, gradient descent.
2. Embeddings, tokenization, next-token prediction.
3. Transformer block: attention, LayerNorm, residuals, MLP.
4. Loss, perplexity, sampling, generation.
5. Attention math, RoPE, KV cache. PLE paper.
6. Fine-tuning, transfer learning, PTQ.
7. Teach one topic to the team.
8. Write the "what I learned" page.

Resources

- Repo: <https://github.com/slvDev/esp32-ai>
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi
- TinyStories paper: <https://arxiv.org/abs/2305.07759>
- Gemma 3n: <https://ai.google.dev/gemma/docs/gemma-3n>
- Attention Is All You Need: <https://arxiv.org/abs/1706.03762>